

Interdisciplinary CS1 Course for Non-Majors: The Case of Graduate Psychology Students

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Abstract—The objective of interdisciplinary education is to teach multiple domains, so that the students can solve integrative problems that require knowledge and skills from these domains. Interdisciplinary programs, in which computer science is combined with other domains, range from bioinformatics to computational linguistics and data science.

Data science is an interdisciplinary field that integrates computer science, mathematics, statistics, and the domain knowledge of the data involved. It is a relevant research skill for many disciplines, including social sciences. Specifically, graduate psychology students who wish to learn data science have already mastered the psychology domain knowledge and their statistics background is advanced. They lack, however, the computer science knowledge required for meaningfully handling data science. The problem is, however, that there is no CS1 course that fits graduate psychology students who study data science: On the one hand, the current CS1 courses do not suit social science students; on the other hand, CS1 courses for non-majors usually do not meet the required level of computer science needed for data science.

Accordingly, to close this gap, we designed a new CS1 course – an interdisciplinary Introduction to Computer Science for Psychological Science course – that a) suits social science students and b) meets the required computer science level for data science. We evaluated the course by examining students' learning outcomes, reflections, and responses on questionnaires. Based on the data analysis according to the self-determination motivation theory, it is concluded that the course design enabled to achieve the course's targets.

Index Terms—Data science, machine learning, computer science education, interdisciplinary, CS1 course, graduate psychology students.

I. INTRODUCTION

Data science is a new field of research that focuses on turning raw data into knowledge and value [1], [2]. It is an interdisciplinary field that integrates knowledge and methods from computer science, mathematics, and statistics, as well as the domain knowledge of the data (see Figure 1). The radical growth in recent years in the availability of both data and the computational resources required to process them has led to a corresponding increase in the demand for data scientists. As a result, new data science education programs are being launched at a growing rate, many of which are offered to undergraduate students [3]. In addition, data science programs are being established on the graduate level in departments of

computer science, statistics, and a variety of other domains, such as social science.

Data science programs require extensive knowledge and skills in mathematics, statistics, and computer science [4]–[7]. Such knowledge is crucial for an understanding and implementation of advanced data science and machine learning applications. While it is possible to teach and learn the principles of data science without knowledge in computer science and programming, using computerized tools such as Orange data science, WEKA, KNIME, and SPSS, such learning is limited.

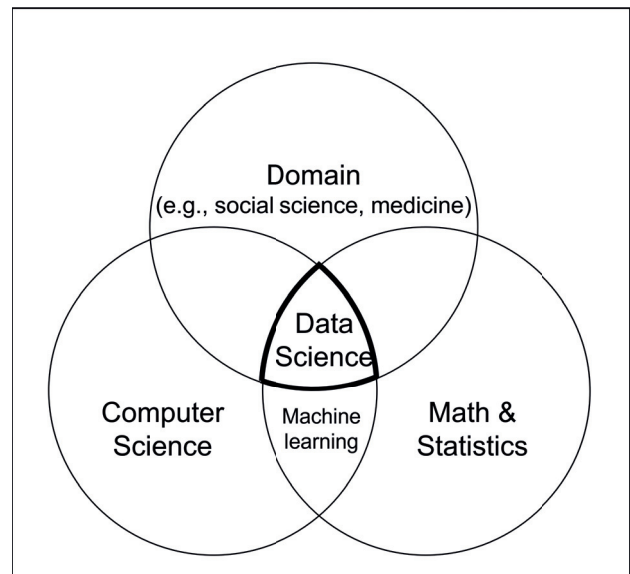


Fig. 1. Data Science is an Interdisciplinary Field. Authors Version, Inspired by Conway [8]

Introduction to computer science and programming courses, often called CS1 [9], are designed to introduce students to the fundamentals of programming. CS1 courses are currently taught on two distinct levels. On the higher level, CS1 courses are intended for computer science majors or non-majors in allied fields such as data science or other sciences, technology, engineering, and mathematics (STEM) [10]. These courses require a deep understanding of computational thinking and algorithmics thinking and set a high bar for students. On the

lower level, CS0 courses [11] or CS1 courses for non-majors [10] aim to expose students to programming and to the essence of computer science; these courses do not expect students to master programming at a high level.

Graduate psychology students who wish to learn data science have already gained considerable psychology domain knowledge and have an advanced background in statistics [12], [13]. They still, however, must acquire the required level of computer science knowledge and skills but may find the learning process of these contents quite challenging. On the one hand, the CS1 course for non-majors is adequate for this population, but its learning goals may not meet the data science requirements; on the other hand, the CS1 courses offered to computer science majors are not appropriate for psychology students as they are taught on a very high level.

To close this gap, we designed a new Introduction to Computer Science course intended for graduate psychology students who wish to learn data science. On the one hand, this new course is suited for social science students, and, on the other hand, it meets the required programming level for data science. The new course is part of a Data Science for Psychological Science specialization offered by Tel Aviv University. As data science is interdisciplinary, all courses in this specialization are interdisciplinary. Accordingly, the course was designed as an interdisciplinary introduction to computer science for psychology majors.

Although CS1 courses for majors have been the focus of a large body of research [14], a similar scope of research on CS1 for non-majors cannot be found. In a recent survey, Becker and Quille [15] mapped only 13 papers on CS1 for non-major courses out of 777 papers on CS1 published by SIGCES in the last 50 years.

According to our literature review (see Section II), the new course design presented in this paper is novel in four aspects:

- Whereas interdisciplinary CS1 courses for STEM majors exist (see, e.g., [16]), it seems that there are no such interdisciplinary courses for non-STEM majors.
- Whereas all existing CS1 courses are intended for undergraduate students, this course is intended for graduate students
- It may be challenging to develop an interdisciplinary course, as an interdisciplinary staff is required [17]. We bypassed this challenge by fostering cooperation between the course staff, who contributed the CS specialty, and the graduate students themselves, who contributed the psychology specialty.
- Working with real-life data is known to motivate students in introductory computer science courses. Whereas in the reviewed research works, non-major students learned using pre-prepared datasets, in our course, the students collected their own data, which were also relevant for their thesis research.

Data for this study were collected from students' learning outcomes, reflections, and questionnaires. The data analysis, conducted according to the self-determination theory [18],

revealed that the interdisciplinary design of the course supports student learning and achievements.

The rest of the paper is organized as follows. In Section II, we review relevant literature regarding data science and computer science education for non-majors (A and B respectively), interdisciplinary education (C), the self-determination theory (D), and the Data Science for Psychological Science course (E). In Section III, we describe our research method, and in Section IV, we present our results. In Section V, we discuss the results, and in Section VI, we conclude.

II. BACKGROUND

In this section, we briefly review the relevant literature regarding the three main components of the interdisciplinary Computer Science for Psychological Science course for graduate psychology students: data science education for non-majors, computer science education for non-majors, and interdisciplinary education. In addition, we present the self-determination theory (SDT), a motivation theory that we used for the data analysis [18] and the Data Science for Psychological Science course on which we focus in this paper.

A. Data Science Education for Non-Majors

New data science education programs are being launched at a growing rate [19]. Most of these programs are designed for undergraduate data science students, and vigorous discussions have been taking place regarding the appropriate curricula of such programs (e.g., [5], [6]). Data science undergraduate programs usually last three to four years and include the teaching of a large body of knowledge from the fields of computer science, mathematics, and statistics.

Several programs were designed as interdisciplinary programs for students majoring in data science and include also the domain knowledge of one or more disciplines [4], [16], [20], [21]. While the need for data science education for non-data science majors is evident [22], there has been very little discussion about the implementation of such programs [23]. Salloum et al. [24] note that "*Currently there is a plethora of M.S. programs... a common complaint by professionals attending these programs is the lack of appropriate preparation (in computer science, math or statistics, depending on the student's background)*" (p. 509).

B. Computer Science Education for Non-Majors

CS1 courses are considered difficult and demanding, and suffer from an overload of learning goals and high failure rates [25]. Although many attribute the high failure rates to the difficult nature of programming, Luxton-Reilly [26] asserted that programming is easy and that the high failure rates are the result of non-realistic learning goals and poor pedagogy.

CS1 courses can be even more difficult for students who are not majoring in computer science and allied disciplines (i.e., data science) and so several curriculum innovations and pedagogical methods have been proposed to support such students [27]–[30]. Forte and Guzdial [28] developed a media computation course, a CS1 course tailored for non-majors, in

which “students learn how different media are encoded electronically and how to write basic code for manipulating digital images, audio, video, and text” (p. 249). A similar initiative was developed by Cortina [27] for engineering students in the form of an introductory computer science course for non-majors that “focuses on the major contributions of computer science from the perspective of the process of computation” (p. 218). The course does not include programming in a programming language but, rather, focuses on algorithms and on the principles of computing and its history. Sullivan [29] proposed a data-centric CS1 course, which covers the basics of databases and data mining and offers an introduction to programming and data visualization.

C. Interdisciplinary Education

Data science is considered an interdisciplinary domain that integrates computer science, mathematics, statistics, and an application domain [3]. The literature describes interdisciplinarity as the cooperation between two or more domains, located, in terms of integration, above multidisciplinary and below transdisciplinarity [31]. In multidisciplinary education, which is the lowest level of integration, learners are expected to gain knowledge and understanding in each domain separately, without being required to apply the knowledge and methods of one domain to solve problems in the other domain. At the highest level of integration, transdisciplinary represents a convergence of two or more disciplines that creates a new, independent discipline.

Interdisciplinarity reflects an integration of two or more scientific fields [32] and represents a higher level of integration than multidisciplinary [33]. Although it is common to find interdisciplinary programs that integrate computer science with other domains, such as biology or neural science, it is not common to find interdisciplinary introduction to computer science courses [14], [15]. The lack of such courses can be explained by the complexity involved in the development and teaching of such courses [17]. For example, the Computer Science in the Biology Laboratory course [16] is an introduction to programming course in which students learn to program and analyze biological data that they collect themselves in a biology laboratory. This course is taught in parallel by two lecturers: one from the computer science domain and one from the biology domain.

Other initiatives reflect the interdisciplinarity of CS1 courses by integrating real-life data from a variety of domains. For example, Burlinson et al. [34] developed BRIDGES, a system that enables the creation of data structure assignments with real-world data and visualizations. Anderson et al. [35] offered a CS1 course in which students learn to connect to accessible databases of real-life data and analyze them. According to a recent literature review, however, these examples are still the exception rather than the rule [14].

D. Self-Determination Theory

The self-determination theory (SDT) is a broad framework for understanding factors that increase intrinsic motivation,

autonomy, and psychological wellness [18]. It emphasizes people’s inherent motivational propensities for learning and growing. Research shows that in educational settings, intrinsic motivation results in positive outcomes [36], [37].

According to SDT, human beings naturally seek growth and wellness, and three primary needs are particularly fundamental for supporting these processes: autonomy, competence, and relatedness. Autonomy concerns a sense of initiative and ownership of one’s actions. It is strengthened by the learner’s sense of interest and value and is weakened by external control, whether by means of rewards or punishments. Competence concerns the feeling of mastery. The need for competence can be fulfilled in education settings by well-structured challenges, positive feedback, and opportunities for growth. Relatedness concerns a sense of belonging and connection. It is enabled by the conveyance of indications of respect and caring.

SDT has been used to evaluate teaching methods in the context of data science and machine learning. For example, Marinescu-Istodor and Jormanainen used SDT to analyze high school pupils’ motivation to learn machine learning [38].

E. Data Science for Psychological Science Course

The Data Science for Psychological Science specialization is a new initiative of the School of Psychological Sciences at the Tel Aviv University. The specialization includes two courses: Introduction to Computer Science for Psychological Science and Introduction to Data Science for Psychological Science. The motivation to offer this specialization is three-fold:

- 1) To expose graduate psychology research students to new research methods that are emerging from the new discipline of data science.
- 2) To expose graduate psychology research students, in general, and cognitive psychology research students, in particular, to state-of-the-art artificial intelligence and machine learning models that are inspired by the human brain and human thinking.
- 3) To address the market demand for data scientists with a confirmed background in social science, in general, and in psychological science, in particular.

The design of the specialization content was inspired by the data science Venn diagram, shown in Figure I. The current psychology curriculum already contains psychology domain knowledge, as well as statistics [12], [13], [39]–[41], and so the graduate psychology students’ understanding of the domain content and the statistics components of data science is relatively high compared with their knowledge and understanding of computer science.

The specialization was divided into two separate courses, each designed as a 52-hour semester course (4 weekly hours): Introduction to Computer Science for Psychological Science and Introduction to Data Science for Psychological Science. The first course was designed to close the computer science knowledge gap and the second course was designed to develop students’ ability to complete the data analysis cycle, that is, to

explore, analyze, model, and predict, using data they collected in the first course.

III. RESEARCH METHOD

A. Research Goal

Due to the lack of CS1 courses designed for non-majors that meet the required programming level for data science, our research goal was to design such a course for graduate psychology students.

B. Method

Accordingly, we designed a new interdisciplinary course, entitled Introduction to Computer Science for Psychological Science. To study the outcome of this course we used the design-based research paradigm [42], [43], which advances design, research and practice concurrently and has demonstrated considerable potential [44]. Design experiments promote both scientific and educational values through the active involvement of researchers in learning and teaching processes. When design-based research is conducted, researchers collaborate with the learners during the research process, designing and implementing interventions systematically to refine the course structure and content, in order to improve its initial design.

C. Course design

The Introduction to Computer Science for Psychological Science course was designed to close the computer science knowledge gap. Accordingly, three learning goals (LG) were defined:

- LG1: To develop students' computational thinking, which is considered an important 21st century skill, and specifically, an important thinking skill required for programming [45].
- LG2: To enable students to collect data for social psychological research using web scraping with Python.
- LG3: To enable students to collect data for cognitive psychological research by means of computerized experiments developed with JavaScript.

Following previous research that showed that integrating domain data can enhance students' engagement with technical content [41], the computer science content and programming tasks were linked to the interests of the psychology graduate students. Specifically, two relevant topics for social psychology students emerged in a focus group facilitated by the psychology school staff members: a) social networks analysis and web scraping, and b) experiments in computational cognition. Interviews with psychology school staff members also revealed that, whereas the common programming environment for social networks analysis and web scraping is Python, experiments in computational cognition are programmed in JavaScript. Website design with HTML constitutes a common body of knowledge required for both web scraping and experiments development.

Accordingly, the following learning topics were included in the course to address the three LGs:

- 1) Computational thinking, 4 hours (LG1)
- 2) Introduction to web scraping, 4 hours (LG2)
- 3) Python programming, 16 hours (LG2)
- 4) Web scraping with Python, 8 hours (LG2)
- 5) JavaScript programming, 12 hours (LG3)
- 6) Programming online psychological experiments with JavaScript, 8 hours (LG3)

Because of the COVID-19 pandemic limitations, the course was designed in a flipped classroom format [46], with asynchronous pre-recorded lectures and weekly online Zoom meetings devoted to answering student questions and working on advanced exercises in small groups.

D. Research Questions

The following research questions were derived from the research goal:

- 1) Does the interdisciplinary CS1 course for psychological science support non-major students' learning needs? If yes, what are the main characteristics that support the learning process? If no, why not?
- 2) Does the interdisciplinary CS1 course for psychology students meet the programming level required for non-major data science students?

E. Research Population

The research participants were 32 students (20 women and 12 men) who attended the Computer Science for Psychological Science course (the first course in the specialization) during the Fall 2020 semester. Approximately two thirds of the students were MA students and a third were PhD students. Approximately half of the students were studying and researching social psychology and the other half were studying and researching cognitive psychology.

Prior to the onset of the first semester of the specialization, we asked the students to report on their motivation to study the course. For this purpose, we formulated a closed questionnaire based on a previous study that we conducted with social science and digital humanities researchers [47]. Eighty-nine percent of the students indicated that they are interested in learning new research tools and 86% indicated that they are interested in learning Python.

We also asked the students to evaluate their pre-course knowledge in the various topics learned in the course, on a 1 to 5 scale (see Figure 2). As expected, the students' background in psychological experiments and statistics was much stronger than their background in computer science.

F. Research Tools

Data were collected using the following tools:

- Introductory questionnaire administered to the students prior to the onset of the semester.
- Mid-semester questionnaire administered to the students half-way through the semester.
- Final questionnaire administered to the students after the end of the semester.

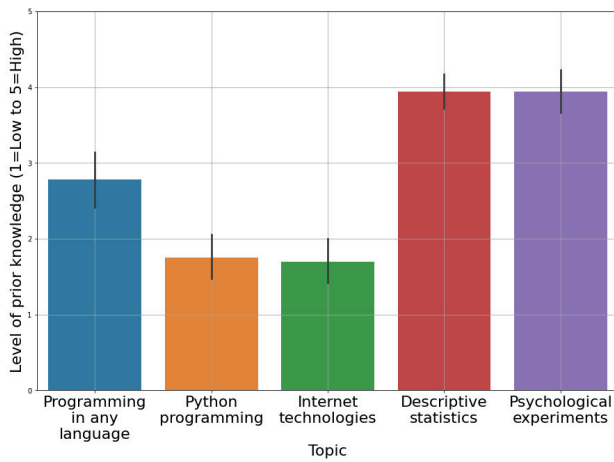


Fig. 2. Students' self evaluation of their prior knowledge (n=36)

- Participant observations of online Zoom lessons taught by the first author of this paper.
- Students' submissions: weekly homework assignments and a final project that implements an individual real research in psychology.

IV. RESULTS

In this section, we analyze the interdisciplinary CS1 course for psychological science from the self-determination theory perspective. The data gathered during the course are presented according to the three fundamental needs that support intrinsic motivation: autonomy, competence, and relatedness.

A. Autonomy

Autonomy refers to a sense of initiative and ownership in one's actions. The course was designed to enhance students' autonomy by two means:

- As the course is taught using hybrid asynchronous lectures and synchronous recitations [48], students had control over the pace of their learning process, as they could decide when and where to watch the pre-recorded lectures.
- Students work individually on weekly homework assignments, as well as on their final project.

Indeed, our data analysis shows that these pedagogical methods meet students' needs for autonomy. In the mid-semester questionnaire, we asked the students to evaluate, on a scale of 1 to 7, the contribution of the different types of educational activities to their learning (see Figure 3). It is notable that individual activities, such as learning from recorded lectures or working on tasks individually, received a higher score than group activities, such as discussions or group tasks. When asked to rank their preferred learning formats with respect to the hybrid learning on a 1-5 scale, their responses indicated that they preferred the asynchronous lectures–synchronous recitation format (see Figure 4), which, once again, enhances their feeling of autonomy in terms of when and where to study the course.

With respect to the course assignments, in a reflective questionnaire distributed to the students after the lesson in which the students presented their projects, two thirds of the students mentioned the following skills that they can apply autonomically at this stage of the course: development of a software tool required for their research; returning to lecture recordings for learning purposes; searching the web for new required features, new required software libraries, and solutions to problems; understanding code examples; and conducting experiments with code.

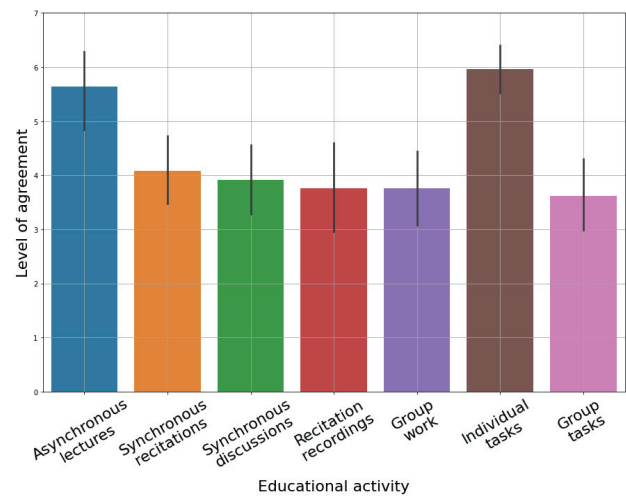


Fig. 3. Students' perception of the contribution of educational activities to their learning (n=25)

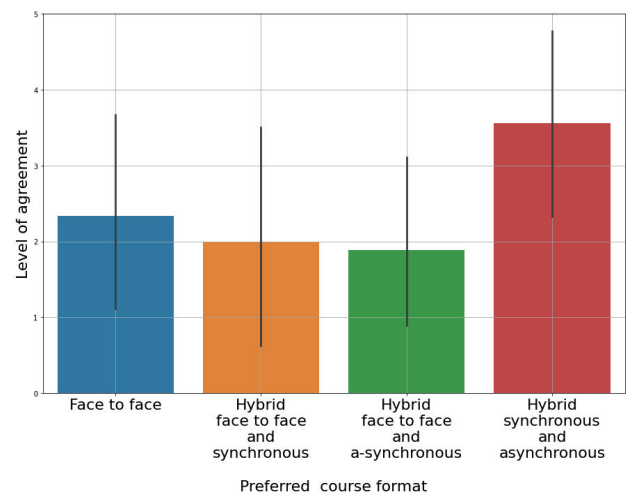


Fig. 4. Students' preferred course format (n=10)

B. Competence

Competence refers to a feeling of mastery. The need for competence can be fulfilled in educational settings by well-structured challenges, positive feedback, and opportunities for growth. While the programming level expected at the end of the course is high, the learning curve was designed for

novice programmers with no prior programming experience. Accordingly, each computational concept and programming technique was taught in a separate short lecture video to enable the students manage their learning time individually during each week for each topic according to their knowledge level, confidence, and process.

In addition, the students received a task on each specific topic which integrated both computer science and data science concepts. For example, during the course, the students developed a system that collects currency exchange rates from the Bank of Israel website. As part of the first and second topics of the course (computational thinking and introduction to web scraping respectively), the students analyzed the HTML structure of the website and wrote an algorithm in a pseudo code that extracts the required information. As part of the third topic on Python programing, the students analyzed the HTML as a string to extract the required information. As part of the fourth topic, web scraping with Python, the students developed automatic code that retrieves the information from the website and analyze it using library tools. Thus, not only was each new topic accompanied by a relevant task, but in addition, students could construct their knowledge gradually which, in turn, further enhanced their sense of competence and increased their motivation.

This last assertion is supported by students' feedback re the course pace. Each week, we asked the students to state whether the course pace was too slow, exactly right, or too fast for them. It is evident that the students perceived the course pace as adequate (see Figure 5), an important component in building students' perception of competence in their process of learning a new skill – programming – and its associated computer science concepts.

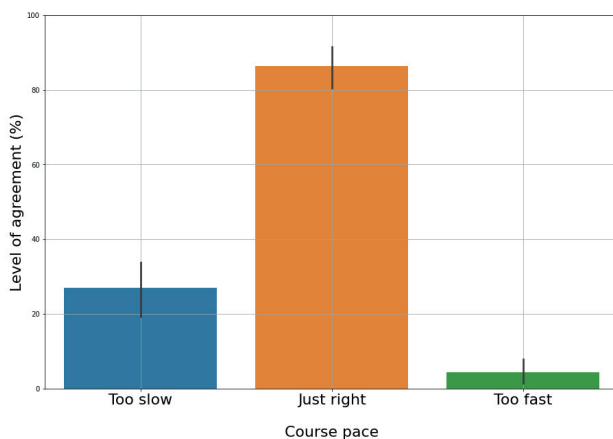


Fig. 5. Students' perception of course pace (n=165 collected by 15 students in 11 questionnaires at the end of each lesson)

We also derived the students' programming competence with respect to the programming requirement for data science from the final project, the students developed as the course final assignment. Here are several examples of the experiments they developed: a cognitive experiment designed to investigate if and to what extent wearing protective masks inhibits

human understanding of facial expressed feelings; a cognitive experiment designed to investigate the relationship between obsessive compulsive disorder (OCD) and indecisive behavior; a cognitive experiment designed to measure visual working memory capacity using the change detection task; a cognitive experiment designed to measure the causal effect of working memory load on outcome-irrelevant reinforcement learning, and a web-scraping project designed to follow the spread of fake news regarding a contagious disease.

The students worked on their projects individually, and two recitations during the semester were devoted to project-related colleague consultation in small groups. Since the projects the students developed were used in their graduate studies research, that is, they accomplished a meaningful task for their real life, the fact that all students completed their projects successfully proves that they reached the required level of programming.

In the end-of-course reflection questionnaire, approximately third of the students reflected on their self-perception of their competence, mentioning that they feel competent to develop programs that will meet their future research needs, as well as to learn new techniques when required. For example, one student wrote *"It is great to realize that I can build experiments on my own and I am sure I will use this ability"*. Thus, based on students' successful project implementation and on students' reflections, we can conclude that the course learning fulfilled their need for competence both with respect to the required programming competence, the self-feeling of mastery in programming, and self-learning capabilities.

C. Relatedness

Relatedness refers to a sense of belonging and connection. The interdisciplinary design of the course and fact that students develop experiments for their own research, based on the material they learn in the course, demonstrate to the students the relatedness of computer science to their own research and thus enhance their feeling of relatedness to both computer science and data science. Approximately one third of the students mentioned relatedness in their reflections on the final project in the end-of-course reflection questionnaire, stating that the programming tools they learned are powerful for the development of their psychological experiments, and that they can use them to collect data that would not otherwise have been accessible for their research. For example, one student wrote *"I think this course is significant for social science students. The course teaches an important research domain, which is rapidly growing these years"*.

V. DISCUSSION

With respect to the first research question "Does the interdisciplinary CS1 course for psychological science support non-major students' learning needs? If yes, what are the main characteristics that support the learning process?", our results indicate that the answer is yes. The main characteristics of the course design enhanced students' internal motivation by fulfilling their needs for autonomy, competence, and relatedness.

With respect to the second research question “Does the interdisciplinary CS1 course for psychology students meet the programming level required for non-data science majors?”, our answer is yes. As the projects developed by the students are also part of their real research conducted as part of their masters or PhD studies, the students’ successful implementation of the data science projects proved that the required level of programming learned in the interdisciplinary CS1 course for non-major data science students was achieved.

VI. CONCLUSION

Integrating real-life data taken from the psychology domain is an important aspect of the interdisciplinary CS1 for psychological science course presented in this paper. This aspect of the course can be generalized beyond the scope of this specific interdisciplinary course. It has been argued that learning computer science with real data can motivate students and support their learning [35]. It may be difficult, however, to design an interdisciplinary computer science course [17]. One method of designing an interdisciplinary computer science course is to simply integrate data from other disciplines [16].

We add that there is a hierarchy of possibilities for integrating real-life data in computer science education (see Figure 6). We suggest that the two highest levels of the hierarchy, both of which use real-life data collected by the students, not only increase student motivation (as is illustrated in this paper), but may also be easier to implement than data (either designed or real) provided by the instructor.

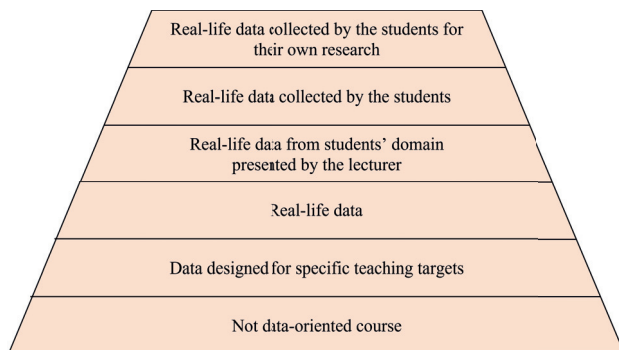


Fig. 6. Proposed range of possibilities for data integration in computer science education)

We plan to continue this research in two ways: First, we plan to continue observing whether graduate psychology students use the course content in their future research, and if yes, how. Second, we intend to investigate new interdisciplinary pedagogies that will enhance the strengths of interdisciplinarity and at the same time, mitigate its challenges.

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